

TECHNOLOGY STACK

Date: 18th July 2026	Project Name: Book a Doctor App	Team Lead: Kilari Ushodaya
Domain: Healthcare	Team Members: Shyam Prasad Mantri, Madhu Hasitha Kondeti, Nanda Kumar Madugula, Mantena Sirisha	

Layer / Subsystem	Selected Technology	Technical Justification & Role
Frontend UI	HTML5, Vanilla CSS3, JavaScript ES6+	Zero heavy dependencies, fast loading, responsive Glassmorphic design flexibility.
Backend Framework	Python 3.10+, Flask REST API	Lightweight, highly scalable REST API microservices framework.
Triage Engine	Python Keyword & Rule Matcher	Parses symptom text inputs to map medical specialty categories.
Database Layer	SQLite (Dev) / PostgreSQL	ACID-compliant transactional integrity for simultaneous booking locks.
Document Vault	Encrypted Storage / AWS S3	Secure storage for diagnostic PDFs and prescription image files.
Notification Gateway	SMTP Email / Twilio SMS / Toast	Delivers booking receipts and automated visit reminder alerts.